

Calvin Hou

Seattle, WA | 206-601-6421 | chou61@gatech.edu | calvinhou.com (Portfolio) | linkedin.com/in/houcalvin

EDUCATION

B.S. in Electrical Engineering, GPA: 3.9

Georgia Institute of Technology

May 2028 (Expected)

Atlanta, GA

TECHNICAL SKILLS

PCB & Embedded Systems: Schematic Design & PCB Layout (Altium, KiCad), Power Electronics, Signal Integrity, Board-Level Debug, STM32, ESP32, CAN FD, UART, 1-Wire

Languages: C, C++, Embedded C, Python

Test & Lab Equipment: Oscilloscope, Logic Analyzer, Multimeter, DAQ Systems, Soldering, Wire Harnessing & Crimping

Coursework: Digital System Design, Programming for Hardware/Software Systems, Signal Processing, Circuits & Electronics

EXPERIENCE

Firmware Research and Design Intern

June 2026 – Aug. 2026

Delta Electronics

Bangkok, Thailand

- Diagnosed EMF-induced signal corruption causing **critical current sensing inconsistencies** in data center PSUs; validated **21** IIR and glitch filter configurations in STM32 embedded C to eliminate erroneous fault readings
- Identified siloed MCAL implementations across Delta's PSU lines and built a standardized library of **6** modules (PWM, ADC, DAC, CAN FD, UART, Watchdog), eliminating redundant firmware rewrites for **8** engineers across **6** product lines
- Led an intern subteam to build Python hardware-in-the-loop test suites across **6** MCAL modules, validating **3,000+** configurations and generating **UL60730-1** certification evidence for regulatory submission

Advanced Robotics Centre Research Intern

Sep. 2024 – Oct. 2024

National University of Singapore

Singapore

- Wired and integrated actuators on a **3-DOF** robotic arm to model and automate labor-intensive port lashing operations
- Implemented an inverse-kinematics algorithm in **C++** for precise XYZ end-effector positioning, achieving **±5mm** repeatability across the full workspace

Hardware Research and Design Intern

Aug. 2024 – Sep. 2024

Asia Vital Components

New Taipei City, Taiwan

- Integrated thermistors with DAQ systems to characterize the unit under a **1kW** dummy heat load; identified airflow inefficiencies and enclosure sealing improvements compiled documentation for the next product revision
- Assembled and pressure-tested CDUs and cooling pumps for a rack-mounted sidecar cooling system targeting **NVIDIA Blackwell** GPU infrastructure in AI data centers

PROJECTS

Avionics Responsible Engineer

Jan. 2026 – Present

YJSP (Yellow Jacket Space Program)

Atlanta, GA

- Designed the STM32-based control PCB for the Control Test Vehicle from schematic to fab in Altium, driving **2x** linear actuators, **4x** RCS fans, and **2x** brushless ESCs
- Designed an ADIN1200 Ethernet PHY and multi-rail power system (22.2V → 12V/4.4V/3.3V) with per-rail ADC telemetry, maintaining signal integrity in a **high-EMI** rocket flight environment

Electrical Systems Member

Aug. 2025 – Present

Hytech Racing (Electric Formula at Georgia Tech)

Atlanta, GA

- Led integration of **6** DS18B20 sensors across battery pack modules (schematic, 1-Wire firmware, on-vehicle validation), achieving **FSAE EM.3.3** compliance
- Introduced a J1772 inlet to the accumulator charging system, enabling **240V** access at competitions and cutting charge time by **50%**

1S USB-C PD Battery Charger | *Altium, ESP32, Embedded C*

Jan. 2026 – Present

- Designed a 4-channel, **15W** 1S Li-ion charger PCB in Altium with thermal and EMI mitigation; wrote ESP32 firmware for an OLED display delivering real-time SOC and thermal monitoring

HONORS & AWARDS

- Amazon Future Engineer Externship, 1st Place, Technical Skills Track**; selected from **3,000+** Seattle-area applicants
- Faculty Honors & Dean's List**, Georgia Institute of Technology
- FIRST Robotics Competition** — Team 9442, Miso Mechanics (Founder, 2023)
 - District Event Winner**; founded the team from scratch, raising **\$45,000+** in sponsorships; ranked **top 4%** globally; fielded **4 robots** with **zero electrical failures**
 - 6 awards**: **2x** Rookie Inspiration, Excellence in Engineering, Industrial Design, Rising All Star, Rookie All Star